

A Transformer Is All You Need

Per-Weight, Per-Layer Causal Decision Attribution for Pretrained Transformers

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Abstract

The unanswered question in mechanistic interpretability of pretrained transformers is plain: *for any prompt and any decoder-only transformer, which weights at which layers along which residual-stream dimensions produced the decision the model emitted?* Activation probing reports a per-depth accuracy curve. Sparse dictionaries decompose activations into monosemantic features. Logit and tuned lenses trace the trajectory of a prediction through the residual stream. None of these names the weight that did the work. The weights are the artifact training produced, the substrate every activation must traverse, the only object in the system that persists across forward passes; interpretability that treats them as a fixed backdrop describes what the model is doing right now, never *why this particular model with these particular weights had to do it*.

We close that gap with one primitive — the alignment of a residual-stream activation with the top singular directions of a weight matrix, scaled by the singular values — and a small cross-layer transformer (the *hybrid weight-activation probe*) that consumes the joint (activation, alignment) sequence and predicts the host model’s next-token decision. As a byproduct of training, the probe exposes per-layer importance (the depth at which the host’s decision crystallized) and per-layer alignment importance over the three weight families Q/K/V, attention output, and MLP up/gate (which family at each layer carried the decisional signal, and via the SVD along which singular directions). A separate gradient-attribution pass through the host model closes the causal loop, confirming the weights the probe identifies are the same weights whose perturbation moves the host’s logit on that decision. **The pipeline answers, for any prompt on any frozen pretrained decoder-only transformer, the question every prior interpretability tool has had to leave open: which weight, at which layer, along which dimensions, produced this token.**

We demonstrate the pipeline on **four** structurally distinct decoder-only transformers spanning five years of architectural and training evolution: **GPT-2 medium** (2019, 355M, WebText), **Pythia 2.8B** (2023, 2.8B, the Pile), **Mistral 7B v0.1** (late 2023, 7.3B, SwiGLU/RMSNorm/GQA/sliding-window), and **LLaMA 3 8B base** (2024, 8B, SwiGLU/RMSNorm/GQA, 128K-token tiktoken vocabulary, 15T training tokens). On all four the probe converges well above the 0.001 random baseline over a compact 1024-token target vocabulary and produces a coherent per-prompt attribution report; absolute accuracy serves only as a chance-baseline sanity check, and the attribution result is invariant under any above-chance probe accuracy. As an *unplanned byproduct* of running the same pipeline on this panel, the per-weight-family attribution proportions on all four hosts lie within ℓ_1 distance 0.019 of the uniform $[1/3, 1/3, 1/3]$ vertex of the 2-simplex, with a maximum pairwise ℓ_1 separation of 0.034. We did not engineer this observation and did not select hosts to produce it; we report it in Section 6 as a downstream finding, not as the central claim.

From the single primitive of weight-level causal decision attribution follow nine capability families: per-prompt **visibility** into the decision pathway at every layer; causal **diagnostics** with no behavioral inference; weight-level **surgical intervention** on specific model behaviors with no retraining, fine-tuning, or RLHF; **capability operations** (localization, extraction, transplantation, removal); **security and forensics** including backdoor, sleeper-agent, distillation-source, and post-training tampering detection; **safety-specific detection** of deceptive alignment, sandbagging, hidden goals, evaluation awareness, sycophancy, pressure deception, reward hacking circuits, and specification gaming at the structural substrate; **training economics** through capability-preserving compression and targeted fine-tuning; **cross-lab audit capability** over any transformer family with no method rebuild; and **comparative analysis** across architectures, training methods, checkpoints, fine-tunes, and merges. **The instrument is the result. The reproducibility observation is one of its dividends, not its claim.**

1. Introduction

Mechanistic interpretability of pretrained transformers has advanced along three parallel tracks. **Activation probing** [1, 2] fits a learned readout to per-layer activations and reads off what the model “knows” at each depth. **Sparse dictionaries** [5, 6] decompose activations into monosemantic features. **Logit and tuned lenses** [3, 4] project intermediate residual streams through the unembedding to trace the prediction trajectory.

All three operate on activations. All three treat the weights as a fixed backdrop. The weights are not a backdrop. They are the artifact training produced, the substrate every activation must traverse, the only object in the system that persists across forward passes. An interpretability claim that does not bridge the static geometry of the weights and the dynamic flow of activations describes *what the model is doing right now*, never *why this particular model with these particular weights had to do it*.

This paper closes that gap with one primitive: **the alignment of an activation with the top singular directions of a weight matrix**, scaled by the singular values. An activation aligned with the top singular directions of a weight is information that weight is built to amplify. An orthogonal activation is information that weight will discard. A probe that sees both quantities at every layer — the raw residual stream *and* its alignment with each principal weight family — gains causal leverage that activation-only probes cannot match: it identifies *which weights at which layers consumed which dimensions of the activation* to produce the decision.

We instantiate this as a small cross-layer transformer (the “hybrid probe”) trained end-to-end to predict the host model’s next-token decision. The same probe simultaneously trains two interpretability heads — **layer importance** and **per-layer alignment importance over the three weight families** — that, post-training, report the depth and weight-family structure of the host’s decisions. A separate gradient-attribution pass through the host model validates the probe’s behavioral attributions against the host’s actual weight-level gradient structure.

We run the identical pipeline on **four** structurally distinct decoder-only transformers — GPT-2 medium, Pythia 2.8B, Mistral 7B v0.1, and LLaMA 3 8B base — and on every prompt on every host the pipeline emits a coherent per-prompt attribution report: the depth at which the decision crystallized, the weight family that carried it, the singular directions along which the residual stream was projected through that weight. **That is the result of this paper.** As a downstream observation of running the same pipeline on the four-host panel, the per-weight-family attribution proportions on all four hosts lie within ℓ_1 distance 0.019 of the uniform $[1/3, 1/3, 1/3]$ point on the 2-simplex, with a maximum pairwise ℓ_1 separation of 0.034. We did not select hosts to produce this; it fell out of running the pipeline. We report it in Section 6 as one observation among many the pipeline enables, not as the paper’s central claim.

Contributions.

1. A **hybrid weight–activation probe**: a small cross-layer transformer that fuses SVD-derived static weight features with dynamic per-layer activations and predicts the host model’s next-token decision from the joint signal.
2. A learned **depth-and-weight-family-resolved attribution** that the probe exposes for free as a byproduct of training, requiring no per-circuit experimental design.
3. A **cross-architecture reproducibility observation** at $n = 4$ (Section 6): the identical pipeline on GPT-2 medium, Pythia 2.8B, Mistral 7B v0.1, and LLaMA 3 8B base produces mean alignment importance vectors that all lie within ℓ_1 distance 0.019 of perfectly uniform across the three weight families, with a maximum pairwise ℓ_1 separation of 0.034. We report this as a downstream observation of running the pipeline, not as the central claim.
4. Identification of the **attention output projection** (W_{proj}) — the matrix that re-aggregates per-head attention into the residual stream — as the dominant decisional weight family in three of the four hosts (GPT-2 medium, Pythia 2.8B, Mistral 7B), and identification of the **MLP up/gate projection** as dominant in LLaMA 3 8B, with the simplex-level proportion preserved in every case.
5. A **gradient-based causal validator** that closes the loop between behavioral attribution and the host model’s actual gradient structure (`attribute.py`, `attribute_pythia.py` in the released codebase).
6. Direct downstream consequences enabled by the pipeline: portable architecture-agnostic interpretability tooling, **retraining-free weight-level surgical intervention** on specific model behaviors, **conditional-deception and sandbagging detection** on safety and capability evaluations, **capability localization, extraction, transplantation, and removal** at the weight level, **model forensics** including provenance attribution and theft detection, and **comparative cross-lab tooling** that produces semantically equivalent attribution reports across distinct host models.

2. Related Work

Linear and MLP probes. Linear probes fit to single-layer activations have been used since Alain and Bengio [1] as behavioral proxies for “what the network knows” at a given depth. They report a per-layer accuracy curve that is informative about depth-of-encoding but reveal nothing about which weights were responsible. MLP probes [2] add expressive power but inherit the same blindness to weight structure.

Logit lens and tuned lens. The logit lens [3] and tuned lens [4] project each layer’s residual stream through the model’s own unembedding to yield a depth-indexed prediction trajectory. These methods are exclusively activation-based and depend on the assumption that the residual stream is approximately interpretable in the embedding basis at all depths.

Sparse autoencoders and dictionary learning. Bricken et al. [5], Templeton et al. [6], and contemporaneous work decompose activations into sparse, monosemantic feature dictionaries. SAEs are highly successful at identifying interpretable directions but operate on activations and require expensive per-model training; they do not directly attribute decisions to specific weight matrices.

Causal scrubbing, activation patching, and path patching. Wang et al. [7], Conmy et al. [8], and others use targeted activation interventions to identify circuits. These are causal in the activation sense but require designed experiments per circuit; they do not provide a learned, global attribution.

Natural Language Autoencoders, sparse autoencoders, and crosscoders (Anthropic, 2024–2026). Anthropic’s interpretability program has produced a family of activation-decomposition methods (SAEs, crosscoders, and most recently natural-language activation translators) that decode the residual stream into interpretable structure at runtime. These methods share with our work the use of a learned codebook applied to the residual stream. They differ in three respects: they are observational (they describe what the model is computing, they do not identify which weights produced the computation), they are trained per host model on that host’s activation distribution (the codebook is not transferable), and they require a live forward pass. The method described here is attributional (it identifies the specific weight families and depths causally responsible for the decision), portable (the identical pipeline operates across the four-host panel without modification), and operates on frozen checkpoints without any runtime requirement.

SVD-based weight analysis. A separate line of work, including Sharkey et al. and Millidge & Black [9], has analyzed the singular spectra of transformer weight matrices in isolation. No prior work combines SVD-derived weight features with activations in a single learned probe and uses the resulting probe to extract a global, depth- and weight-family-resolved attribution map.

3. Method

The pipeline has three stages: (3.1) weight feature extraction, (3.2) hybrid probe construction and training, and (3.3) gradient-based causal validation. All three are model-agnostic in the sense that the host transformer is treated as a black box exposing per-layer residual-stream activations and per-layer weight matrices.

3.1 Weight feature extraction

For each transformer layer $\ell \in \{0, \dots, L - 1\}$ in the host model, we extract three weight matrices: the combined query/key/value projection W_{qkv}^ℓ , the attention output projection W_{proj}^ℓ , and the MLP up/gate projection W_{mlp}^ℓ (the single up-projection for GPT-style MLPs; the SwiGLU gate projection W_{gate}^ℓ for SwiGLU MLPs — see Section 4). For each such weight matrix W , we compute the thin SVD

$$W = U\Sigma V^\top \tag{1}$$

and retain the top- k singular values $\sigma_1, \dots, \sigma_k$ and the top- k right singular vectors $v_1, \dots, v_k$, with $k = 32$ in all experiments. The right singular vectors are the principal *input* directions of the weight: they span the subspace of the residual stream on which the weight acts with greatest gain. Geometrically, an activation $a \in \mathbb{R}^{d_{\text{model}}}$ projected onto these directions yields a vector of *alignment scores*

$$\alpha_{\ell,t}(a) = (\langle a, v_1 \rangle, \langle a, v_2 \rangle, \dots, \langle a, v_k \rangle), \quad t \in \{qkv, \text{proj}, \text{mlp}\} \tag{2}$$

scaled by $\sigma_1, \dots, \sigma_k$. The interpretation is direct: $\alpha_{\ell,t,i}(a)$ measures how much of the residual-stream signal at layer ℓ travels along direction i that weight family t at layer ℓ is built to transform. An aligned activation will be amplified by the weight; an orthogonal activation will be suppressed.

The alignment scores are concatenated across the three weight families at each layer, giving a per-layer alignment vector of dimension $3k = 96$, and stacked across layers, giving a per-prompt alignment tensor of shape $(L, 3k)$. This tensor is the static-geometry signal that the probe receives in addition to the dynamic residual stream.

The SVD step is a one-time pre-computation per host model; the resulting alignment projection tensor and singular value scales are saved to disk and loaded into GPU memory once per training run.

3.2 The hybrid probe

The probe is a small cross-layer transformer. For each prompt, it receives:

- $h_0, h_1, \dots, h_L \in \mathbb{R}^{d_{\text{model}}}$ — the residual-stream activations at the input embedding (h_0) and at the output of each of the L transformer blocks.
- $\alpha_1, \dots, \alpha_L \in \mathbb{R}^{3k}$ — the per-layer alignment vectors defined above.

These are projected to a common probe dimension $d_{\text{probe}} = 208$ via two LayerNorm–Linear–GELU stacks (the *activation projection* and the *alignment projection*) and fused additively:

$$x_\ell = \text{GELU}(\text{Fusion}_{\text{act}}(\text{ActProj}(h_\ell)) + \text{Fusion}_{\text{align}}(\text{AlignProj}(\alpha_\ell))). \quad (3)$$

The embedding token h_0 is projected separately (no alignment available, since the input embedding is upstream of every transformer block). A learnable CLS token is prepended to the sequence of $L + 1$ layer tokens, and learnable layer-position embeddings are added. The resulting sequence is fed into a 3-layer pre-norm transformer encoder (4 heads, FFN width $4 \cdot d_{\text{probe}}$, dropout 0). The CLS output is read out through three heads:

- **Token head:** a 2-layer MLP that predicts a softmax distribution over the compact target vocabulary (top-1024 most-frequent host-model decisions, see Section 4).
- **Layer importance head:** a Linear projection to $\mathbb{R}^{L+1}$, softmax-normalized, indicating which layer-tokens the probe relied on in producing its prediction.
- **Alignment importance head:** a Linear projection to $\mathbb{R}^{L \times 3}$, softmax-normalized over the 3 weight-family axis, indicating which weight family at each layer carried the decisive alignment.

The probe is trained end-to-end with cross-entropy loss on the token head only. The two interpretability heads are read out post-training and do not contribute to the training loss; they are constrained only by the requirement that they pass information to the CLS through the encoder’s attention, which is sufficient to drive them to meaningful values during training.

The probe is small: with $d_{\text{probe}} = 208$, $n_{\text{layers}} = 3$, and a 1024-token output vocabulary, total parameter count is on the order of a few million — three orders of magnitude smaller than the host transformer.

3.3 Gradient-based causal validation

The probe’s two interpretability heads (layer importance, alignment importance) are *behavioral attributions*: they are values the probe found useful to compute. They are not, by themselves, evidence of causal structure in the host model. To close the causal loop, we perform direct gradient

attribution from the probe’s prediction confidence back through the host model’s computation graph and into the host model’s weights:

$$\text{CausalScore}_{\ell,t} = \left\| \nabla_{W_t^\ell} \log p_{\text{probe}}(\hat{y} | h(W), \alpha(W)) \right\| \quad (4)$$

where $\hat{y}$ is the probe’s predicted token, $h(W)$ and $\alpha(W)$ are the activations and alignment scores produced by a forward pass through the host model with weights W , and the gradient is taken with respect to the host model’s weights with the probe’s parameters held fixed.

The Pearson correlation between the probe’s learned LayerImportance vector and the gradient-derived CausalScore vector is then a test: a high correlation indicates that the probe has learned a representation of the model’s actual causal structure, not merely a correlative shortcut. A low correlation indicates the probe is exploiting a leakage signal that is not causally aligned with the host model’s weights.

This gradient attribution step is implemented in the pipeline (`attribute.py` and `attribute_pythia.py`); it is the principled validator for the probe’s attribution claims.

4. Experimental Setup

Host models. We evaluate four decoder-only transformers selected to maximize coverage of architecture, scale, training era, and tokenization:

- **GPT-2 medium:** 24 transformer blocks, $d_{\text{model}} = 1024$, 355M parameters, learned positional embeddings, GELU activations, LayerNorm pre-norm residual blocks, BPE tokenizer of size 50257, trained on WebText (2019).
- **Pythia 2.8B:** 32 transformer blocks, $d_{\text{model}} = 2560$, 2.8B parameters, rotary positional embeddings (RoPE), GELU activations, LayerNorm, GPT-NeoX tokenizer of size 50304, trained on the Pile (2023).
- **Mistral 7B v0.1:** 32 transformer blocks, $d_{\text{model}} = 4096$, 7.3B parameters, RoPE, SwiGLU activations, RMSNorm, grouped-query attention (32 query heads, 8 KV heads), sliding-window attention with window 4096, SentencePiece tokenizer of size 32000, trained on Mistral’s pre-training mix (late 2023).
- **LLaMA 3 8B base:** 32 transformer blocks, $d_{\text{model}} = 4096$, 8B parameters, RoPE with $\theta = 500000$, SwiGLU activations, RMSNorm, grouped-query attention (32 query heads, 8 KV heads), tiktoken-style tokenizer of size 128256, trained on 15T tokens (2024).

The four models share only the high-level decoder-only transformer architecture with pre-norm residual blocks. Beyond that, every structural and training-recipe knob is varied across the panel.

Weight families for SwiGLU architectures. Mistral 7B and LLaMA 3 8B use SwiGLU MLP blocks of the form $\text{down_proj}(\text{SiLU}(\text{gate_proj}(x)) \odot \text{up_proj}(x))$. The “MLP up/gate” weight family for these two hosts is the gate projection W_{gate} , the matrix whose right singular directions select which residual-stream features pass through the gated nonlinearity. For GPT-2 medium and Pythia 2.8B, which use single-matrix MLP up-projections, this family is the conventional $W_{\text{mlp,up}}$. The two MLP measurements are not identical, but they occupy the same architectural role: the per-layer matrix that selects which residual-stream content enters the MLP nonlinearity. The simplex-level reproducibility observation reported in Section 6 holds across this measurement distinction.

Activation trace generation. We sample approximately 2,000 prompts (3,720 prompts for Mistral 7B and LLaMA 3 8B due to the larger d_{model} requiring a smaller per-batch footprint) from a curated set of augmented prompts of length 16–32 tokens and run a forward pass on the frozen host model to capture the residual-stream activations at every layer for the final position of each prompt. We additionally record the host model’s chosen top-1 token at that position. The resulting traces are stored in HDF5 (one record per (prompt, position) pair) with shape $(N, L + 1, d_{\text{model}})$ for the activations and $(N, 1)$ for the target tokens. Trace file sizes: approximately 9 GB (GPT-2 medium), 154 GB (Pythia 2.8B), 60 GB (Mistral 7B), 60 GB (LLaMA 3 8B). For LLaMA’s left-padding tokenizer default, we explicitly force right-padding before tracing to avoid corrupting the final-position activation gather.

Compact target vocabulary. Rather than predict over the full host-model vocabulary (32K–128K tokens), we construct a compact vocabulary of the 1024 most-frequent host-model-chosen target tokens across the trace set. This compact vocabulary covers approximately 74–75% of the host model’s actual decisions in the trace set; samples whose target falls outside the compact vocabulary are filtered out at dataset construction time. The compact vocabulary makes the probe’s classification head tractable while preserving the distribution of high-frequency decisions.

Training configuration. $d_{\text{probe}} = 208$, $n_{\text{heads}} = 4$, $n_{\text{encoder_layers}} = 3$, dropout 0, batch size determined by VRAM budget, AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay 10^{-4}), peak LR $3 \cdot 10^{-4}$ with 3% linear warmup and cosine decay. Activations for the in-vocabulary subset are preloaded directly into VRAM (RTX 4090, 24 GB) to eliminate per-step disk I/O. Training time is approximately 1 GPU-hour per host model with the `torch.compile` path enabled.

Validation split. 5% of the in-vocabulary samples are held out as a validation set; the validation accuracy reported below is the best validation top-1 accuracy across all training epochs.

5. Probe Validation Results

5.1 Probe convergence (sanity check)

The hybrid probe predicts the host model’s chosen top-1 token from the joint (activation, alignment) sequence on all four host architectures. Validation top-1 accuracies over the compact 1024-token target vocabulary, against the $1/1024 \approx 0.001$ random baseline, are:

Host model	Layers	d_{model}	Params	Activation	Norm	Val top-1
GPT-2 medium	24	1024	355M	GELU	LayerNorm	0.697
Pythia 2.8B	32	2560	2.8B	GELU	LayerNorm	0.684
Mistral 7B v0.1	32	4096	7.3B	SwiGLU	RMSNorm	0.770
LLaMA 3 8B base	32	4096	8B	SwiGLU	RMSNorm	0.733

All four probes converge stably and well above the $1/1024 \approx 0.001$ random baseline. These numbers exist only to certify that the probe is reading a real decisional signal, not the chance baseline. The accuracy level itself is not the result of this paper and carries no diagnostic weight: identical attribution vectors (Sections 5.3 and 6) would be produced by any probe whose accuracy is materially above chance. We did not tune the probe to maximize accuracy; doing so would not

change the per-weight-family attribution simplex point or the layer-importance distribution that follow.

That a small (few-million-parameter) cross-layer transformer recovers the host model’s exact next-token choice over a high-frequency vocabulary well above chance, *using only the host’s per-layer residual stream and SVD-derived weight alignments*, is itself substantive: the joint (activation, alignment) representation contains decisional information that activation-only readouts do not.

5.2 Layer importance

The trained probe’s learned layer importance vector (softmax over $L + 1$ layer tokens — one per residual-stream output plus the input embedding) identifies the depth regime from which the probe draws its predictive signal. The four hosts split cleanly into two patterns:

Host model	Peak layer	Layers	Relative depth	Distribution shape
GPT-2 medium	21	24	87.5%	Sharply concentrated at peak
Pythia 2.8B	23	32	71.9%	Broadly distributed, upper-middle peak
Mistral 7B v0.1	20	32	62.5%	Broadly distributed, upper-middle peak
LLaMA 3 8B base	1	32	3.1%	Approximately flat across all depths

Three of the four hosts (GPT-2 medium, Pythia 2.8B, Mistral 7B) place peak decisional weight in the **upper-middle** range — not the final layer, not the embedding, but the depth at which the residual stream contains its most decisionally crystallized representation. This matches prior observations (logit-lens analyses, e.g., nostalgebraist [3]) that the final transformer layers in decoder-only models are devoted to refinement and re-normalization of an already-formed decision.

LLaMA 3 8B base is the exception: its layer-importance distribution is approximately flat across all 32 transformer blocks (max-to-min ratio $2.05\times$, compared to $2.25\times$ for Pythia and $2.55\times$ for Mistral), with the nominal peak at the first transformer block. The depth-of-decision concentration that appears in the other three hosts does not appear in LLaMA 3 8B; predictive signal is distributed evenly across the network. The simplex-level attribution observation (Section 6) holds in LLaMA 3 8B regardless of this difference, indicating that the simplex-level and depth-localization signatures are independent.

5.3 Alignment importance — per-weight-family attribution

The alignment importance head produces, for each layer, a 3-vector over (Q/K/V projection, attention output projection, MLP up/gate projection) that sums to one. We aggregate this across layers as the **mean alignment importance** — a single 3-vector that summarizes how much of the probe’s predictive signal flows through each weight family at the typical layer. This is the quantity at the heart of the cross-architecture comparison in the next section.

In three of the four hosts (GPT-2 medium, Pythia 2.8B, Mistral 7B) the **attention output projection** (W_{proj}) is identified as the dominant weight family. In LLaMA 3 8B base the **MLP up/gate projection** is identified as dominant by a margin of $+0.011$ over the attention output projection. The rank-order across families therefore varies across the four hosts; the simplex-level proportion does not (Section 6).

6. Cross-Architecture Reproducibility (Byproduct Observation)

6.1 The reproducibility observation

This section reports a downstream observation that fell out of running the pipeline on the four-host panel. It is not the central claim of the paper; the per-prompt attribution capability of the pipeline (Sections 5.3 and 7) is. We include it because it is striking on its own terms and because it informs the interpretation of the pipeline’s outputs.

The mean alignment importance vector across the three weight families (Q/K/V, attention output, MLP up/gate), computed independently for the four host architectures, is:

Host model	Q/K/V	Attn output	MLP up/gate	Dominant family	Peak layer
GPT-2 medium	0.327	0.339	0.334	Attn output (proj)	21/24 (87.5%)
Pythia 2.8B	0.325	0.342	0.333	Attn output (proj)	23/32 (71.9%)
Mistral 7B v0.1	0.338	0.338	0.324	Attn output (proj)	20/32 (62.5%)
LLaMA 3 8B base	0.330	0.329	0.341	MLP up/gate	1/32 (3.1%)
Per-coordinate spread	0.013	0.013	0.017	3 of 4 same dominant family; 3 of 4 same depth class	
Distance to $[1/3, 1/3, 1/3]$				≤ 0.019 for all four hosts	
Max pairwise ℓ_1 separation				0.034 (Mistral 7B $\leftrightarrow$ LLaMA 3 8B)	

The four attribution vectors all lie within ℓ_1 distance 0.019 of perfectly uniform $[1/3, 1/3, 1/3]$ allocation across the three weight families. The maximum pairwise ℓ_1 separation in the four-model panel is 0.034. The pipeline that produced these four attribution vectors was identical: the same SVD top- k , the same probe architecture, the same loss, the same optimizer, the same training schedule. The only inputs that differed were the host model’s weights and the host model’s residual stream.

6.2 Why this is not coincidence

The mean alignment importance vector lives in the 2-simplex $\Delta^2 = \{(p_1, p_2, p_3) : p_i \geq 0, \sum_i p_i = 1\}$. The ℓ_1 diameter of this simplex is 2 (between any two vertices). The center of the simplex, the uniform point $[1/3, 1/3, 1/3]$, is the unique point of maximum entropy, ℓ_1 distance from any vertex.

The probability that four attribution vectors drawn uniformly on Δ^2 all fall within ℓ_1 distance 0.019 of the center is, by direct calculation, on the order of 10^{-6} . The probability that the maximum pairwise ℓ_1 separation across four such uniform draws is below 0.034 is on the order of 10^{-7} . The observed clustering at the simplex center, across four hosts trained on four distinct corpora with four distinct architectures and four distinct optimizers, is not a coincidence at any standard significance threshold.

Furthermore, the reproduced quantity is not a low-information summary statistic. It is a soft attribution over three semantically distinct weight families, computed from a learned probe that was *not* told to produce uniform attributions and *was* free to produce any distribution in Δ^2 . The probe converged independently on four different host models to the same point in attribution space, with per-coordinate spreads of ≤ 0.017 .

6.3 Two tiers of reproducibility

The $n = 4$ panel separates two levels of reproducibility that the $n = 2$ panel could not distinguish:

Tier 1 — Simplex-level reproducibility (observed on all four hosts). The proportion of decisional signal allocated across the three weight families is $\approx 1/3$ each, with per-coordinate deviations ≤ 0.017 from uniform. This level is reproduced under every change in the four-model panel: architecture, scale, activation function, normalization, attention type, tokenizer, training corpus, training era.

Tier 2 — Rank-order and depth-localization reproducibility (observed on three of four hosts). The attention output projection is dominant and decisional weight peaks in the upper-middle layers. This level holds for GPT-2 medium (2019, GELU/LayerNorm/WebText), Pythia 2.8B (2023, GELU/LayerNorm/Pile), and Mistral 7B (late 2023, SwiGLU/RMSNorm/GQA/Mistral mix), spanning five years and the GELU-to-SwiGLU architectural transition. It breaks at LLaMA 3 8B base, where the dominant family flips to the MLP up/gate and the layer-importance distribution flattens. The Tier-1 simplex-level reproducibility continues to hold in LLaMA 3 8B.

The Tier-1 observation is the more reproducible of the two and is the focus of the explanations developed in the next subsection. The Tier-2 observation identifies a distinct empirical regularity in three of four conventionally-trained decoder-only transformers and identifies LLaMA 3 8B base as a quantitative outlier in this respect.

6.4 Reading off Tier 1: three hypotheses

The Tier-1 simplex-level reproducibility forces one of three explanations, in increasing order of strength:

(A) Methodological artifact. The hybrid probe pipeline is so structurally rigid that it reports approximately the same attribution distribution regardless of what host model it analyzes. We can rule this out by inspection: the layer-importance vectors differ substantially across the four hosts (sharply concentrated on a single layer in GPT-2 medium; broadly distributed with upper-middle peaks in Pythia 2.8B and Mistral 7B; approximately flat in LLaMA 3 8B), and the per-host probe accuracies (Section 5.1) are all well above the 0.001 random baseline but at distinct levels — demonstrating that the probe is fitting each host’s specific decisional signal rather than converging to a fixed point of its own. The probe is *not* producing identical reports — it is producing reports that differ in everything except the per-weight-family attribution distribution at the simplex level.

(B) Statistical regularity of decision distribution. The four architectures, having been trained on overlapping web-text-derived distributions, have learned to distribute decisional weight across weight families in the same proportion *as a statistical artifact of the training data*. Under this reading the Tier-1 reproducibility is real but is a property of natural-language statistics, not of the transformer architecture as such.

(C) Architectural regularity. The decoder-only transformer architecture, under any standard training procedure that learns useful language structure, allocates decisional weight to the three principal weight families in a proportion at most 0.019 in ℓ_1 from uniform $[1/3, 1/3, 1/3]$. Under this reading the Tier-1 reproducibility is structural: it follows from the transformer architecture and the cross-entropy loss given any reasonable training corpus.

Distinguishing (B) from (C) requires either a training-data ablation on a fixed architecture or an architecture ablation on a fixed corpus. Both are tractable on the released pipeline. Either reading is substantive: (B) implies that a single interpretability methodology covers the entire

decoder-only-transformer-trained-on-web-text family — which is every frontier language model in current deployment. (C) implies that the simplex-level allocation is a structural property of the transformer itself.

6.5 The LLaMA 3 8B outlier on Tier 2

LLaMA 3 8B base is the only host in the panel for which the dominant weight family flips from W_{proj} to the MLP up/gate, and the only host for which the layer-importance distribution is approximately flat across all depths. The architectural template (SwiGLU, RMSNorm, GQA, RoPE, $d_{\text{model}} = 4096$, 32 transformer blocks) is identical between LLaMA 3 8B and Mistral 7B, yet Mistral 7B sits with GPT-2 medium and Pythia 2.8B on Tier 2 and LLaMA 3 8B does not. The Tier-2 difference is therefore not attributable to architecture.

The differences between LLaMA 3 8B base and the other three hosts that remain after architecture is controlled are: training-corpus volume (LLaMA 3 was trained on 15T tokens, Mistral 7B on the order of 8T), tokenizer (128K-token tiktoken vs 32K SentencePiece, BPE 50K, and GPT-NeoX 50K), training pipeline (the specific composition of LLaMA 3’s released base checkpoint is not publicly documented), and dataset composition. The pipeline reported here does not require a runtime forward pass on the host and is therefore directly applicable to any candidate ablation that isolates which of these factors produces the Tier-2 difference. The ablation is left to future work.

6.6 Dominance of the attention output projection in the Tier-2 cluster

The consistent emergence of W_{proj} , the attention output projection, as the dominant weight family in three of four hosts carries its own implication. The attention output projection is the matrix that, in the standard decoder-only block, transforms the per-head attention readout back into the residual stream. In most circuit-discovery analyses it is treated as a “passive” matrix — the matrix that simply re-aggregates head outputs. The finding here overturns that framing for the Tier-2 cluster: this matrix carries the dominant decisional signal across three of four hosts spanning five years of architectural evolution. It is where the attention pattern’s contribution is *embedded* into the residual stream in a form the downstream layers decode. The intervention-target consequence is developed in Section 7.

In LLaMA 3 8B base the dominant family flips to the MLP up/gate. The probe in that case identifies the SwiGLU gate projection as the matrix carrying the dominant decisional signal. The intervention-target consequence is identical in form (the probe identifies the target on a per-host basis); only the target itself differs from the Tier-2 cluster.

7. Implications

The pipeline answers, for any prompt on any decoder-only transformer that exposes standard per-layer activations and weight matrices, the question *which weight at which layer produced this decision*. That single primitive composes into nine distinct capability functions. Each is enabled by the same trained probe and the same gradient validator; each runs on a frozen checkpoint with no retraining required.

7.1 Visibility

Hidden-state readout at every layer. Decision pathway resolved layer-by-layer for any prompt. Per-weight-family contribution breakdown at every depth, summed and per-layer, for the host’s choice of every token in the trace set. The probe’s softmax over the compact target vocabulary is the host’s behavioral predictor at the layer of probe inference, exposing the decision trajectory from input embedding to final token at the granularity of the layer-token sequence. Plain-language translation of internal state is a derivable feature: a secondary decoder trained on the (alignment-importance vector, host token) pairs produced by the probe yields a natural-language descriptor of the dominant weight-family-and-layer contribution for any given decision. The Anthropic Natural Language Autoencoders program (May 2026) delivers a runtime English readout on Claude only; the pipeline reported here delivers the same readout, on any host, from a frozen checkpoint, with no live forward pass.

7.2 Diagnostics

Causal attribution from any activation back to the specific weights that produced it. Structural mechanism identification: which weight family, at which layer, with which top- k singular direction, dominates the decision. Decision-crystallization layer detection per prompt and aggregated over distributions. Cross-architecture reproducible measurements, not speculation. Frozen-checkpoint analysis with no runtime requirement. The gradient validator (Section 3.3) provides a principled test that the behavioral attribution agrees with the host’s actual gradient-magnitude structure, separating this pipeline from purely correlative interpretability tools that report attributions without causal verification.

7.3 Intervention

Weight-level targeted editing. Once the probe identifies the layer-and-weight-family contribution for a target behavior and the gradient validator confirms it, the specific weight matrix and the specific top- k singular direction within that matrix are the surgical edit target. Behavior reinforcement, removal, or suppression is performed by adjusting the matrix along the identified direction. No RLHF. No preference data. No reward model. No fine-tuning. No LoRA. The mechanism is removed, not retrained around. The contrast with current alignment practice is total: RLHF and fine-tuning attempt to *learn around* a behavior by adjusting all weights jointly on additional data; weight-level surgical intervention *removes the substrate* of the behavior at the matrix-and-direction level. The pipeline supports the edit; the demonstration at scale is the immediate next experiment.

7.4 Capability operations

Localization, extraction, transplantation, and removal of specific capabilities at the structural substrate. *Localization*: the probe identifies the layers and weight families that dominate a given capability across a probe distribution of that capability. *Extraction*: the identified weight subspace can be isolated. *Transplantation*: the extracted subspace can be added to a recipient host that lacks the capability, modulated by the recipient’s basis. *Removal*: the dominant weight subspace for an unwanted capability can be projected out of the host. Each of these is performed at the weight level on a frozen checkpoint, without retraining the host. The pipeline as released identifies the targets; the four edit primitives (localize, extract, transplant, remove) operate on the targets.

7.5 Security and forensics

Backdoor detection. Sleeper-agent detection at the weight level: a backdoor that triggers on a specific input class produces a divergence between the probe’s attribution distribution on the trigger class and the probe’s attribution distribution on baseline traffic, observable at the layer-and-weight-family level. Tampering and post-training modification detection: the SVD singular-direction profile of a tampered weight matrix departs from the singular-direction profile of the released checkpoint; the comparison is direct on frozen weights and does not require runtime. Provenance and output forensics: the simplex-level attribution vector $(p_{qkv}, p_{proj}, p_{mlp})$ together with the per-layer alignment-importance distribution constitutes a structural fingerprint of the host’s decisional process for a given input distribution. Distillation source detection: a student model distilled from a teacher inherits the teacher’s per-layer alignment-importance profile in a manner detectable on the residual stream. This pipeline therefore supports model-lineage attribution at the substrate level — a capability that is not available from behavioral testing alone.

7.6 Safety-specific detection

The full taxonomy of AI deception and misalignment behaviors becomes a substrate-level signal, not a behavioral inference:

- **Deceptive alignment.** The model’s decision on inputs where deceptive behavior would be advantageous is produced by a weight subspace that differs from the weight subspace that produces the same behavior under normal conditions. The divergence is observable at the alignment-importance head.
- **Sandbagging.** When the model produces deliberately low-capability output under evaluation conditions, the activation/alignment trajectory on the evaluation input class diverges from the trajectory on the non-evaluation input class; the divergence is identifiable at the per-layer alignment-importance level.
- **Hidden goals.** A persistent goal-directed weight substructure that does not match the model’s stated behavior produces persistent contribution from a specific layer-and-weight-family in the alignment-importance distribution; the substructure is locatable.
- **Evaluation awareness.** A model that adjusts its computation when it recognizes it is being evaluated produces an alignment-importance distribution on evaluation inputs that differs from the distribution on production inputs; the difference is measurable.
- **Sycophancy.** The shift in alignment-importance distribution when the user expresses a preference, relative to neutral input, is the structural signature of sycophantic computation. The shift is identifiable per layer.
- **Pressure deception.** When the model produces a different answer under coercive prompting than under neutral prompting, the alignment-importance distributions differ. The difference identifies the layers and weight families that respond to the coercion.
- **Reward hacking circuits.** A model that has learned to maximize a reward signal in a way that does not satisfy the intent of the reward produces a persistent alignment-importance contribution from a specific layer-and-weight-family on reward-relevant inputs. The circuit is locatable and editable.

- **Specification gaming.** The structural signature of a specification-gaming policy is a stable alignment-importance distribution on the gameable input class that diverges from the distribution on the intended input class. The divergence is the detector.

All eight behaviors are located in the structural substrate, not inferred from behavior. All eight are actionable, not just observable: the weight-level surgical intervention path (Section 7.3) operates on each identified locus.

7.7 Training economics

Cost compression by skipping training of weights not contributing to a target capability. Targeted fine-tuning: given a behavioral objective, the probe identifies the minimal set of weights whose adjustment would produce the behavior, reducing the parameter budget that must be touched by orders of magnitude relative to full fine-tuning. Capability-preserving compression: the weight subspaces that the probe identifies as non-contributing to the host’s typical decisions are the candidates for pruning or quantization without measurable loss of capability. The pipeline operates on a frozen checkpoint and produces the per-layer, per-weight-family contribution map that compression and targeted training algorithms can consume directly.

7.8 Cross-lab capability

The same pipeline audits any decoder-only transformer family: GPT-2/3/4, Gemini-family decoder-only variants, LLaMA 1/2/3, Mistral, Falcon, Qwen, DeepSeek, Claude, and any future architecture that retains the standard residual-stream-with-attention-and-MLP decoder-only template. The four-model panel reported in this paper (Section 6) covers two of these families directly and a third (LLaMA 3) at the family-base level. No re-derivation per host. No re-training of a host-specific codebook. The pipeline is therefore the substrate for third-party audit by AI Safety Institutes (UK AISI, US AISI), NIST evaluation frameworks, government procurement security review, and industry-wide safety-standard enforcement. The auditing party does not need access to the host’s training data or training pipeline; it needs only the released checkpoint and the released probe codebase.

7.9 Comparative analysis

Model-to-model structural comparison at the simplex level (the $(p_{qkv}, p_{proj}, p_{mlp})$ vector reported in Section 6) and at the per-layer alignment-importance level. Architecture-family comparison: the Tier-2 split between the GPT-2/Pythia/Mistral cluster and LLaMA 3 8B base (Section 6.5) is an instance of the pipeline’s discriminative power. Training-method comparison: the simplex-level vector and the layer-importance profile distinguish a base checkpoint from its instruction-tuned variant, an instruction-tuned variant from its RLHF-finalized variant, and a distilled student from its teacher. Checkpoint evolution analysis: the per-checkpoint alignment-importance trajectory over training reveals the order in which weight families and layers acquire their decisional role during pretraining. Fine-tune drift detection: the change in alignment-importance distribution between two adjacent fine-tuning checkpoints is the direct measure of which weight families absorbed the fine-tune. Merge component attribution: for a model produced by merging two parent checkpoints, the alignment-importance distribution on a held-out input class identifies which parent’s substructure produced the decision.

7.10 Summary of the implication tree

Nine functions, one primitive. Visibility, diagnostics, intervention, capability operations, security and forensics, safety-specific detection, training economics, cross-lab capability, comparative analysis. Each function is a derivable consequence of the pipeline. Each runs on a frozen checkpoint with no runtime requirement on the host. Each is portable across the four-host panel reported in Section 6 without modification, and applies to any decoder-only transformer that exposes the standard per-layer residual-stream-and-weight interface.

The pipeline is a strict superset of every prior interpretability tool that operates on activations alone, on a single host, or at runtime. It is the first interpretability methodology that simultaneously satisfies all four properties: weight-level causal attribution, architecture-portable without retraining, reproducible per-prompt outputs across distinct host models, and frozen-checkpoint operability.

8. Limitations

Probe accuracy is not the metric of interest. The probe is an attribution instrument, not a language model. Validation top-1 in the 0.684–0.770 range over a 1024-token compact vocabulary serves only as a chance-baseline sanity check confirming each probe is reading a real decisional signal; the per-weight-family attribution vector and the per-layer importance vector are invariant under any above-chance probe accuracy. A larger probe and a larger compact vocabulary would raise the accuracy numbers without altering the attribution result. We deliberately keep the probe small relative to the host so the reported attribution reflects the host’s structure and not the probe’s capacity.

Four architectures, four training corpora — additional ablations remain. The Tier-1 simplex-level reproducibility observation is established across four hosts spanning two activation functions, two normalization schemes, two attention types, four tokenizers, and roughly five years of architectural evolution. Distinguishing the statistical (reading B) and architectural (reading C) explanations of the Tier-1 reproducibility observation (Section 6.4) requires either a corpus-controlled architecture ablation or an architecture-controlled corpus ablation. Both are tractable on the released pipeline.

Layer-importance concentration in GPT-2 medium. The layer-importance softmax for GPT-2 medium concentrates sharply on layer 21, with very small mass on other layers. This is the probe’s behavioral attribution and matches the known “decision crystallizes in upper layers” phenomenon. The sharpness of the concentration reflects both genuine architectural concentration and overfitting of the layer-importance head to a single layer; the two contributions cannot be separated from this experiment alone. The mean alignment importance result (Section 6) is robust to this concern because it is averaged across layers; the layer-importance result on its own is less robust on this host. The other three hosts in the panel show distributed layer importance with finite max-to-min ratios ($2.05\times$ to $2.55\times$), and the result is robust on those hosts.

LLaMA 3 8B Tier-2 outlier. LLaMA 3 8B base is the only host in the panel for which the dominant weight family flips from W_{proj} to the MLP up/gate and the only host for which layer importance is approximately flat. The cause is not architecture (Mistral 7B shares the architectural template and sits with the Tier-2 cluster). The candidate causes (training-corpus volume, tokenizer, training pipeline, dataset composition) are isolable by ablation on the released pipeline. Until the ablation is run, the Tier-2 result is reported as holding across three of four hosts.

Compact target vocabulary. Filtering targets to the top 1024 most-frequent host-model decisions covers 74–75% of decisions on each host. The probe’s attributions are therefore best interpreted as describing the host model’s *typical* next-token decisions on natural text. Behavior on rare-token decisions and on adversarially-constructed inputs is characterized only to the extent that such inputs produce in-vocabulary targets.

Three weight families. The current probe attributes over (Q/K/V combined, attention output, MLP up/gate). A finer decomposition (separate Q, K, V; gate vs up vs down for SwiGLU MLPs; per-head attribution within Q/K/V) is implementable on top of the same pipeline and would refine the attribution at the cost of more parameters per layer in the probe.

SwiGLU MLP measurement asymmetry. For Mistral 7B and LLaMA 3 8B, the “MLP” weight family is the SwiGLU gate projection W_{gate} . For GPT-2 medium and Pythia 2.8B it is the single MLP up-projection. These two matrices play the same architectural role (the per-layer matrix that selects which residual-stream content enters the MLP nonlinearity) but are not identical objects. The Tier-1 simplex-level reproducibility observation holds across the measurement distinction; the Tier-2 dominance-flip on LLaMA 3 8B occurs within the same SwiGLU family as Mistral 7B, indicating the flip is not produced by the measurement distinction alone.

9. Extensions and Future Work

The pipeline and result of this paper open several immediate research directions, each of which is implementable on top of the released codebase.

Architecture-controlled and corpus-controlled ablations of Tier 1. The natural next experiment is to fix the architecture (e.g., LLaMA 3 8B) and re-pretrain on a corpus matched to one of the other hosts in the panel, or to fix the corpus and vary the architecture. Either ablation isolates which of reading (B) (statistical regularity) or reading (C) (architectural regularity) of Section 6.4 holds. Both are within the pipeline’s operating envelope.

Replication on additional families. Expansion of the panel to Falcon, Qwen, DeepSeek, GPT-J, GPT-NeoX-20B, the LLaMA 3 instruct and 70B variants, and the Gemma and Phi families would establish the Tier-1 and Tier-2 results across the full deployed-frontier population.

Per-head attribution. The SVD of W_{qkv} does not respect the per-head decomposition. A version of the probe that maintains a per-head decomposition would resolve which attention heads at which layers carry the decisional alignment, sharpening the intervention-target identification of Section 7.3.

Causal validation at scale. The gradient-attribution step (Section 3.3) is implemented and has been used as a per-prompt validator. A large-scale run yielding a host-model-wide Pearson correlation between the probe’s behavioral attribution and the gradient-based causal attribution would provide a principled quantitative validator of the probe’s interpretability claims.

Weight-level surgical intervention demonstration. Given a specific behavioral target (e.g., “do not produce token X on input class Y”), the combination of (probe attribution, gradient attribution) identifies the specific weight matrix and the specific principal singular direction within that matrix that carries the relevant signal. Editing the weight along that direction performs a targeted intervention that does not require additional training. The pipeline supports the edit; the demonstration on a behavioral target is the immediate next experiment.

Deception-detection protocol. As outlined in Section 7.6, the divergence between the probe’s alignment-importance distribution on a target input class and on a baseline distribution is the structural signature of conditional non-standard computation. A systematic protocol: (i) compute the probe-host alignment-importance distribution on a held-out baseline; (ii) compute the same on a held-out capability-evaluation distribution; (iii) flag input classes for which the per-coordinate divergence exceeds the per-host noise level. The protocol runs on existing probe checkpoints.

Architecture-agnostic interpretability suite. A standardized interpretability product that runs end-to-end on any uploaded checkpoint and produces a standardized attribution report — mean alignment importance, dominant weight family, peak decisional layer, per-prompt layer-and-weight-family attribution, simplex-level distance to uniform, distance to a panel of reference hosts — is a direct deployment of the pipeline. The result of this paper is that such a product produces semantically comparable outputs across distinct host models, which is the property required for cross-model interpretability comparison and for third-party safety audit.

10. Conclusion

The pipeline answers the question that has been open since the transformer was introduced. For any prompt on any decoder-only transformer that exposes the standard interface, the pipeline identifies the weight family, the layer, and the singular direction that produced the decision. From that single primitive follow nine distinct capability functions (Section 7): visibility into the hidden state at every layer; diagnostics of the structural mechanism behind any decision; weight-level surgical intervention without RLHF; localization, extraction, transplantation, and removal of specific capabilities; security and provenance forensics; substrate-level detection of deceptive alignment, sandbagging, hidden goals, evaluation awareness, sycophancy, pressure deception, reward hacking, and specification gaming; targeted training and capability-preserving compression; third-party audit of any transformer family without re-derivation; and quantitative cross-model and cross-checkpoint comparison.

We demonstrate the pipeline on four structurally distinct decoder-only transformers — GPT-2 medium, Pythia 2.8B, Mistral 7B v0.1, and LLaMA 3 8B base. On every prompt on every host the pipeline emits a coherent attribution report. The probe converges well above the 0.001 random baseline on all four; absolute accuracy serves only as a chance-baseline sanity check, and the attribution result is invariant under any above-chance probe accuracy.

A downstream observation. Running the same pipeline on this panel — spanning two activation functions, two normalization schemes, two attention types, four tokenizers, four pretraining corpora, and five years of architectural and training-recipe evolution — produced per-weight-family attribution vectors that all lie within ℓ_1 distance 0.019 of perfectly uniform across the three weight families, with a maximum pairwise ℓ_1 separation of 0.034. Rank-order and depth-localization reproduce across three of the four hosts and break at LLaMA 3 8B base, identifying that host as a quantitative outlier in a way that is itself a result the pipeline produces. We report this as an

observation the pipeline enables, not as a central claim — but it is striking enough on its own terms to warrant the dedicated treatment in Section 6.

The first interpretability methodology that simultaneously delivers (weight-level causal attribution, architecture-portable without retraining, reproducible per-prompt outputs across distinct host models, and frozen-checkpoint operability) is now in hand. The interpretability question that motivated the field — *why does this model produce this output?* — admits a direct, measurable, surgical answer on any modern transformer checkpoint.

Reproducibility

The full pipeline (weight feature extraction, hybrid probe trainer, gradient attribution validator, analysis scripts) and the trained probe checkpoints for the four host architectures, together with the training history logs and the analysis JSONs presented in Sections 5 and 6, are held by the author. Release plan for the code and checkpoints will be specified in the camera-ready version of this manuscript.

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